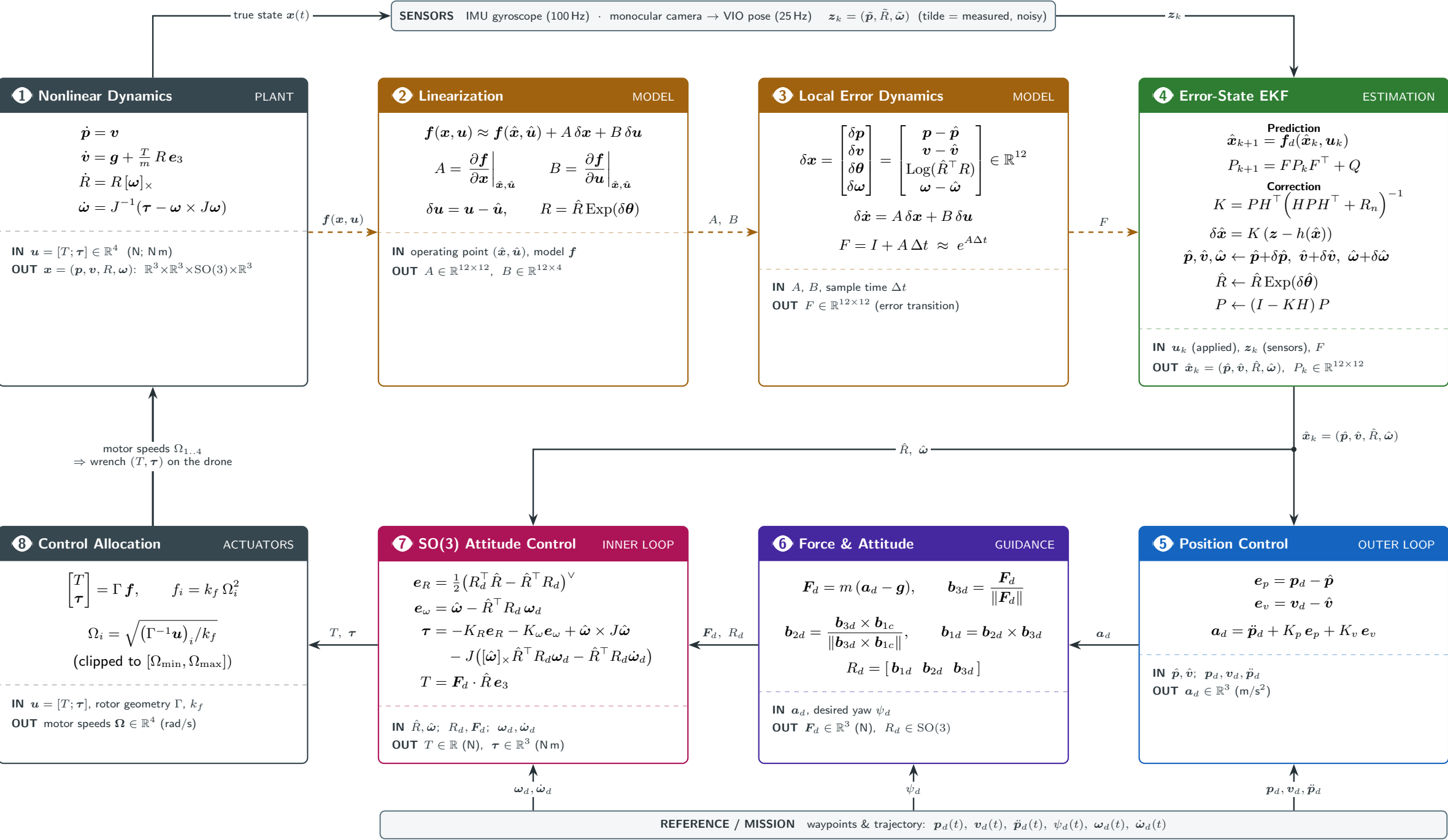


Model 1 — Nonlinear Model-Based Estimation and Geometric SO(3) Control

Closed loop running on the drone: plant → sensors → EKF → outer loop → inner loop → motors → plant



→ runtime signal (every sample k) - - - - - model information used to build the filter ENU world frame (z up), FLU body frame; hat $\hat{\cdot}$ = estimate, subscript d = desired

Notation and dimensions

States, inputs, parameters				Linearization and SO(3) tools				Estimation (EKF)				Control and actuation			
$\boldsymbol{p}$	position (world, ENU)	$\mathbb{R}^3$	m	$[\boldsymbol{a}]_{\times}$	skew matrix, $[\boldsymbol{a}]_{\times} \boldsymbol{b} = \boldsymbol{a} \times \boldsymbol{b}$	$\mathbb{R}^{3 \times 3}$	–	$\hat{\boldsymbol{x}}$	estimated state ($\hat{\boldsymbol{p}}, \hat{\boldsymbol{v}}, \hat{\boldsymbol{R}}, \hat{\boldsymbol{\omega}}$)	18 nos.	mixed	$\boldsymbol{p}_d, \boldsymbol{v}_d, \ddot{\boldsymbol{p}}_d$	desired pos./vel./acc.	$\mathbb{R}^3$	m, m/s, m/s ²
$\boldsymbol{v}$	velocity (world)	$\mathbb{R}^3$	m/s	$(\cdot)^{\vee}$	inverse of $[\cdot]_{\times}$	$\mathbb{R}^3$	–	Δt	EKF / IMU sample time	$\mathbb{R}$	s (0.01)	$\boldsymbol{e}_p, \boldsymbol{e}_v$	position / velocity error	$\mathbb{R}^3$	m, m/s
$\boldsymbol{R}$	attitude, body → world	SO(3)	–	Exp, Log	rot. vector ↔ rot. matrix	$\mathbb{R}^3 \leftrightarrow \text{SO}(3)$	rad	P_k	error covariance	$\mathbb{R}^{12 \times 12}$	m ² , rad ² , ...	K_p, K_v	position / velocity gains	$\mathbb{R}^{3 \times 3}$	s ^{–2} , s ^{–1}
$\boldsymbol{\omega}$	body angular velocity	$\mathbb{R}^3$	rad/s	$(\cdot)$	estimate (EKF output)	–	–	Q	process-noise covariance	$\mathbb{R}^{12 \times 12}$	as P	$\boldsymbol{a}_d$	desired acceleration	$\mathbb{R}^3$	m/s ²
T	total thrust (body z)	$\mathbb{R}$	N	$\hat{\boldsymbol{x}}, \hat{\boldsymbol{u}}$	operating point	18, $\mathbb{R}^4$	mixed	$\boldsymbol{z}$	measurement (cam: 6, gyro: 3)	$\mathbb{R}^m$	m, rad, rad/s	$\boldsymbol{F}_d$	desired force	$\mathbb{R}^3$	N
$\boldsymbol{\tau}$	body torque	$\mathbb{R}^3$	N m	$\delta \boldsymbol{x}$	error state $[\delta \boldsymbol{p}; \delta \boldsymbol{v}; \delta \boldsymbol{\theta}; \delta \boldsymbol{\omega}]$	$\mathbb{R}^{12}$	m, m/s, rad, rad/s	$h(\cdot)$	measurement model	$\mathbb{R}^{18} \rightarrow \mathbb{R}^m$	–	$\psi_d, \boldsymbol{b}_{1c}$	desired yaw, heading	$\mathbb{R}, \mathbb{R}^3$	rad, –
$\boldsymbol{u}$	input $[T; \boldsymbol{\tau}]$	$\mathbb{R}^4$	N, N m	$\delta \boldsymbol{\theta}$	attitude error, $\boldsymbol{R} = \hat{\boldsymbol{R}} \text{Exp}(\delta \boldsymbol{\theta})$	$\mathbb{R}^3$	rad	$\boldsymbol{H}$	measurement Jacobian	$\mathbb{R}^{m \times 12}$	–	$\boldsymbol{b}_{1d}, \boldsymbol{b}_{2d}, \boldsymbol{b}_{3d}$	desired body axes (unit)	$\mathbb{R}^3$	–
$\boldsymbol{x}$	state ($\boldsymbol{p}, \boldsymbol{v}, \boldsymbol{R}, \boldsymbol{\omega}$)	12 DOF	mixed	$\delta \boldsymbol{u}$	input error $\boldsymbol{u} - \hat{\boldsymbol{u}}$	$\mathbb{R}^4$	N, N m	R_n	measurement-noise covariance	$\mathbb{R}^{m \times m}$	as $\boldsymbol{z}^2$	$\boldsymbol{R}_d$	desired attitude	SO(3)	–
m	mass	$\mathbb{R}$	kg (1.5)	$\boldsymbol{f}$	dynamics $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$	12	mixed	K	Kalman gain	$\mathbb{R}^{12 \times m}$	mixed	$\boldsymbol{e}_R, \boldsymbol{e}_{\omega}$	attitude / rate error	$\mathbb{R}^3$	–, rad/s
$\boldsymbol{J}$	inertia diag(0.02, 0.02, 0.04)	$\mathbb{R}^{3 \times 3}$	kg m ²	$\boldsymbol{A}$	state Jacobian $\partial \boldsymbol{f} / \partial \boldsymbol{x}$	$\mathbb{R}^{12 \times 12}$	mixed	$\boldsymbol{f}_d$	one discrete step of $\boldsymbol{f}$	18 → 18	mixed	$\boldsymbol{\omega}_d, \dot{\boldsymbol{\omega}}_d$	desired rate, rate derivative	$\mathbb{R}^3$	rad/s, rad/s ²
$\boldsymbol{g}$	gravity $[0, 0, -9.81]^{\top}$	$\mathbb{R}^3$	m/s ²	$\boldsymbol{B}$	input Jacobian $\partial \boldsymbol{f} / \partial \boldsymbol{u}$	$\mathbb{R}^{12 \times 4}$	mixed	$\delta \hat{\boldsymbol{x}}$	correction computed by the EKF	$\mathbb{R}^{12}$	mixed	K_R, K_{ω}	attitude / rate gains	$\mathbb{R}^{3 \times 3}$	N m, N m s
$\boldsymbol{e}_3$	unit vector $[0, 0, 1]^{\top}$	$\mathbb{R}^3$	–	$\boldsymbol{F}$	discrete error transition	$\mathbb{R}^{12 \times 12}$	–	$\boldsymbol{I}$	identity matrix	$\mathbb{R}^{12 \times 12}$	–	$\Gamma, \boldsymbol{f}, k_f$	mixer, rotor thrusts, thrust const.	$\mathbb{R}^{4 \times 4}, \mathbb{R}^4, \mathbb{R}$	–/m, N, N s ²
$(\cdot)_k$	value at $t_k = k \Delta t$	–	–									$\boldsymbol{\Omega}$	rotor speeds	$\mathbb{R}^4$	rad/s